

Chapter 1 – Introduction

1.1 Background

The rapid growth of digital technologies, cloud computing, interconnected systems, Internet of Things (IoT) devices, and online services has significantly increased the dependence of individuals and organizations on computer networks and information systems. Along with these technological developments, the cybersecurity landscape has become increasingly complex. Cyber attackers continuously develop new techniques to exploit vulnerabilities, compromise systems, steal sensitive information, disrupt services, and conduct financial or politically motivated attacks. Malware, phishing, ransomware, distributed denial-of-service (DDoS) attacks, exploitation of vulnerabilities, and credential-based attacks are among the major threats faced by modern organizations.

The increasing volume, velocity, and complexity of cyber threats have made traditional security approaches insufficient for identifying and responding to threats in a timely manner. Security systems generate large quantities of logs, alerts, incident reports, vulnerability information, threat reports, and other forms of security-related data. A considerable portion of this information is unstructured or semi-structured textual data obtained from security blogs, incident reports, vulnerability databases, security advisories, online forums, news articles, and other sources. Manually examining such information requires significant time and specialized expertise and may result in delays or inconsistencies in threat analysis.

Cyber Threat Intelligence (CTI) has emerged as an important component of modern cybersecurity. CTI involves the collection, processing, analysis, and dissemination of information related to cyber threats and threat actors. The objective of CTI is not simply to collect security information but to transform available data into meaningful intelligence that can support security monitoring, threat detection, incident response, risk assessment, and decision-making.

However, the continuously increasing amount of CTI data presents a major analytical challenge. Cybersecurity analysts must identify relevant information, determine the type of threat, extract important indicators, understand relationships among events, and recognize changes in threat patterns. Automated techniques based on Machine Learning (ML), Natural Language Processing (NLP), and topic modeling can assist in processing large-scale textual CTI data and extracting useful patterns.

Threat classification is one important task in automated CTI analysis. Classification systems can categorize threat-related information into predefined classes such as malware, phishing, ransomware, DDoS, vulnerability exploitation, and other attack categories. Automated classification can reduce the effort required for manual analysis and enable security teams to organize and prioritize large amounts of threat information.

Another important challenge is the detection of emerging threats. Existing classification systems generally depend on predefined categories and labeled datasets. Consequently, they may have difficulty identifying new or evolving threats that do not fit existing categories. Emerging threats may initially appear as changes in the frequency, terminology, entities, techniques, or topics discussed across CTI sources. Temporal topic modeling provides a way to analyze such changes by examining how topics evolve over time.

Therefore, this research focuses on developing a framework for **real-time cyber threat classification and emerging threat detection using Machine Learning and temporal topic modeling**. The proposed research combines supervised threat classification with temporal analysis of cyber threat-related textual information. The objective is to support automated identification of known threat categories while also identifying changing or emerging threat patterns from CTI data.

1.2 Cyber Threat Intelligence

Cyber Threat Intelligence (CTI) is the process of collecting, processing, analyzing, and communicating information about cyber threats in a form that can support cybersecurity decisions and actions. CTI enables organizations to move beyond the simple detection of individual security events by providing contextual information about threats, threat actors, attack techniques, vulnerabilities, and potential impacts.

Modern CTI is generated from multiple sources, including security reports, vulnerability databases, malware analysis reports, security advisories, incident reports, online forums, blogs, news articles, and other publicly available or organizational sources. A significant amount of this information is textual and unstructured, making automated processing an important research area.

The CTI lifecycle generally involves several interconnected activities:

1. **Planning and Direction** – defining intelligence requirements and determining what information is required.
2. **Collection** – gathering relevant threat information from appropriate sources.
3. **Processing** – cleaning, normalizing, structuring, and preparing collected information for analysis.
4. **Analysis** – extracting meaningful patterns, relationships, indicators, and threat information.
5. **Dissemination** – delivering analyzed intelligence to relevant stakeholders.
6. **Feedback** – evaluating the usefulness of intelligence and refining future collection and analysis activities.

Machine Learning and NLP techniques can support multiple stages of this lifecycle, particularly processing and analysis. Automated models can identify patterns in large datasets that may be difficult to discover through manual analysis.

1.2.1 Definition of Cyber Threat Intelligence

Cyber Threat Intelligence can be defined as evidence-based knowledge regarding existing or potential cyber threats that is collected, processed, analyzed, and communicated to support informed cybersecurity actions. Unlike raw security data, CTI provides context and interpretation that can help security professionals understand the nature, source, behavior, and potential consequences of cyber threats.

CTI may contain information about threat actors, malware families, attack techniques, vulnerabilities, indicators of compromise (IoCs), targeted organizations, affected technologies, and observed attack behaviors. The usefulness of CTI depends on the quality, timeliness, relevance, and accuracy of the information.

For automated cyber threat analysis, textual CTI reports are particularly valuable because they contain descriptions of attacks, vulnerabilities, techniques, tools, threat actors, and observed behaviors. Extracting useful information from these reports requires appropriate NLP and ML techniques.

1.2.2 Types of Cyber Threat Intelligence

Cyber Threat Intelligence is commonly categorized into four major types: strategic, tactical, operational, and technical intelligence.

Strategic Cyber Threat Intelligence

Strategic CTI provides high-level information intended primarily for decision-makers and organizational leadership. It focuses on broader trends, risks, threat landscapes, potential business impacts, and long-term cybersecurity considerations.

Strategic intelligence can assist organizations in understanding how changes in the cyber threat environment may affect business operations and security investments.

Tactical Cyber Threat Intelligence

Tactical CTI focuses on the techniques, tactics, and procedures used by threat actors. It helps security professionals understand how attackers operate and which defensive controls may be required.

Examples include information about attack techniques, exploitation methods, malware behavior, and attacker procedures.

Operational Cyber Threat Intelligence

Operational CTI provides information about ongoing or planned cyber campaigns and attacks. It may include information about threat actors, campaigns, targeted sectors, attack timelines, and observed activities.

Operational intelligence can support incident response and security operations by providing contextual information about active threats.

Technical Cyber Threat Intelligence

Technical CTI focuses on technical indicators associated with cyber attacks. Examples include IP addresses, domain names, URLs, file hashes, malware signatures, email addresses, and other Indicators of Compromise (IoCs).

Technical CTI is frequently used by security tools such as Security Information and Event Management (SIEM), intrusion detection systems, endpoint security platforms, and threat intelligence platforms.

The four types of CTI are complementary. Strategic intelligence supports long-term decisions, tactical intelligence describes attacker techniques, operational intelligence provides information about campaigns, and technical intelligence provides actionable technical indicators.

1.3 Problem Statement

The increasing volume of cyber threat information has created significant challenges for cybersecurity analysts and organizations. Large quantities of CTI are continuously generated from security reports, vulnerability disclosures, incident analyses, online sources, and other threat-related information channels. Much of this information is unstructured textual data, making manual analysis time-consuming and difficult to scale.

Existing approaches to cyber threat analysis face several challenges.

1.3.1 Large Volume of Unstructured CTI Data

Cyber threat information is generated continuously and often contains lengthy textual descriptions. Manually reading and analyzing large numbers of reports requires substantial time and human effort.

1.3.2 Difficulty in Automated Threat Classification

Threat reports may describe different attack types using varying terminology and writing styles. A classification system must identify meaningful textual patterns to accurately categorize threat information.

1.3.3 Dependence on Predefined Threat Categories

Many supervised classification approaches require predefined labels. Such systems are effective for known threat categories but may have difficulty handling previously unseen or evolving threat patterns.

1.3.4 Difficulty in Detecting Emerging Threats

New cyber threats can emerge gradually through changes in attack techniques, terminology, tools, vulnerabilities, and threat-actor behavior. Conventional classification systems may not explicitly model these temporal changes.

1.3.5 Temporal Changes in Cyber Threat Information

The characteristics of cyber threats change over time. A topic that is relatively uncommon in one period may become significantly more prominent later. Therefore, analyzing CTI without considering its temporal dimension may result in the loss of important information about emerging trends.

1.3.6 Need for Timely Analysis

Cybersecurity operations require timely identification of threats. Delayed processing of threat information can reduce the practical value of intelligence. Automated approaches are therefore required to process and analyze threat information efficiently.

Based on these challenges, there is a need for an automated framework that can classify known cyber threats while also analyzing temporal changes in CTI data to identify potentially emerging threat patterns.

1.4 Research Motivation

The primary motivation of this research is the increasing complexity and continuously changing nature of the cyber threat landscape. Cyber attacks evolve rapidly, and new vulnerabilities, malware families, attack techniques, and campaigns can emerge over time. Security teams therefore require automated methods that can process large quantities of threat intelligence and identify meaningful patterns efficiently.

Traditional manual analysis is difficult to scale because analysts must examine large volumes of heterogeneous information. Machine Learning provides an opportunity to automate the classification of threat-related information based on patterns learned from historical data.

However, classification alone may not be sufficient for identifying new or evolving threats. A threat classification model generally operates according to categories present in its training data. Emerging threats may initially appear as previously unseen combinations of terms, entities, techniques, or topics. This creates a need for complementary analytical techniques that can examine changes in threat-related information over time.

Temporal topic modeling can provide such complementary analysis by identifying topics within textual CTI data and examining how their distributions change across different time periods. A sudden increase or significant change in a topic may provide an indication of an emerging threat pattern that requires further investigation.

The combination of supervised classification and temporal topic analysis therefore provides the motivation for this research. The proposed approach aims to support both **known threat identification** and **emerging threat pattern analysis** within a unified CTI framework.

1.5 Research Objectives

The main objective of this research is to develop a framework for automated cyber threat classification and emerging threat detection using Machine Learning and temporal topic modeling.

The specific objectives are:

1. **To collect and prepare cyber threat intelligence data** from relevant threat reports and textual CTI sources.
 2. **To preprocess CTI textual data** using appropriate Natural Language Processing techniques such as text normalization, tokenization, stop-word removal, and feature preparation.
 3. **To develop Machine Learning-based classification models** for automatically categorizing cyber threat information into predefined threat classes.
 4. **To compare the performance of different classification algorithms** using appropriate evaluation metrics such as accuracy, precision, recall, and F1-score.
 5. **To apply topic modeling techniques** to identify important topics and thematic patterns within cyber threat intelligence data.
 6. **To incorporate temporal analysis into topic modeling** to investigate how threat-related topics change across different time periods.
 7. **To identify potentially emerging threat patterns** by analyzing changes in topic distributions, frequencies, and temporal trends.
 8. **To develop an integrated framework** combining threat classification and temporal topic analysis for automated CTI analysis.
 9. **To evaluate the effectiveness of the proposed framework** using appropriate experimental datasets and performance measures.
 10. **To analyze limitations and potential improvements** for future development of automated cyber threat intelligence systems.
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1.6 Research Scope

The scope of this research is focused on the automated analysis of textual Cyber Threat Intelligence using Machine Learning, Natural Language Processing, and temporal topic modeling.

The research includes the following areas:

- Collection and preparation of textual cyber threat intelligence data.
- Text preprocessing and normalization.
- Feature extraction and text representation.
- Machine Learning-based cyber threat classification.
- Comparative evaluation of classification algorithms.
- Topic modeling of CTI documents.
- Temporal analysis of extracted topics.
- Identification of changing and potentially emerging threat patterns.
- Evaluation using classification and topic-analysis metrics.
- Development of an integrated analytical framework for CTI.

The research primarily focuses on **textual CTI reports and threat-related documents**. The framework is intended to analyze information contained within the available dataset rather than replace operational cybersecurity systems.

The research does not attempt to provide a complete enterprise Security Operations Center (SOC), automated incident-response platform, or offensive cybersecurity system. It also does not claim that every newly detected temporal topic represents a confirmed zero-day vulnerability or previously unknown cyber attack. Emerging-topic detection is treated as an analytical mechanism for identifying patterns that may require further cybersecurity investigation.

The effectiveness of the proposed framework will depend on factors such as dataset quality, availability of sufficient labeled data, class distribution, language characteristics, temporal coverage, and the quality of extracted textual features.

1.7 Contributions

The major contributions of this research are as follows:

1.7.1 Automated Cyber Threat Classification

The research develops an automated Machine Learning-based approach for categorizing textual CTI into predefined cyber threat classes. This reduces the dependence on completely manual classification of large collections of threat reports.

1.7.2 Comparative Evaluation of Machine Learning Models

Multiple Machine Learning approaches are evaluated using consistent experimental conditions and standard classification metrics. The comparison provides an understanding of how different models perform on the selected CTI dataset.

1.7.3 Temporal Topic Analysis of CTI

The research incorporates temporal analysis into topic modeling to investigate how cyber threat topics evolve across different time periods. This provides an additional perspective beyond conventional document-level classification.

1.7.4 Emerging Threat Pattern Identification

The proposed framework analyzes changes in topic distributions and temporal trends to identify potentially emerging threat patterns. Such patterns can be used as signals for further cybersecurity analysis.

1.7.5 Integrated CTI Analysis Framework

The research combines supervised threat classification with temporal topic modeling into an integrated analytical framework. This enables the analysis of both known threat categories and changing threat-related topics.

1.7.6 Experimental Analysis

The proposed approach is evaluated using quantitative metrics and comparative experiments. The analysis includes classification performance, confusion matrices, precision, recall, F1-score, topic distributions, temporal trends, and emerging-threat analysis.

1.7.7 Reproducible Research Workflow

The research follows a structured data-processing and experimentation workflow covering data preparation, model training, evaluation, topic modeling, temporal analysis, and visualization. This provides a reproducible foundation for further research in automated CTI analysis.

1.8 Thesis Organization

This thesis is organized into the following chapters:

Chapter 1 – Introduction

This chapter introduces the research background and provides an overview of Cyber Threat Intelligence. It discusses the problem statement, research motivation, objectives, scope, major contributions, and organization of the thesis.

Chapter 2 – Literature Review

This chapter presents a detailed review of existing research related to Cyber Threat Intelligence, Machine Learning-based cyber threat classification, Natural Language Processing, topic modeling, temporal

analysis, and emerging cyber threat detection. The chapter also identifies limitations and research gaps in existing approaches.

Chapter 3 – Research Design and Data Preparation

This chapter describes the research design, data sources, dataset characteristics, data collection process, dataset construction, preprocessing procedures, feature representation, and experimental data preparation.

Chapter 4 – Proposed Methodology

This chapter presents the proposed framework for real-time cyber threat classification and emerging threat detection. It describes the overall architecture, Machine Learning classification pipeline, topic modeling approach, temporal analysis, emerging-threat detection mechanism, and evaluation methodology.

Chapter 5 – Experimental Results and Analysis

This chapter presents the experimental setup and results obtained from the proposed methodology. It includes dataset distribution analysis, model implementation, classification performance, comparative model analysis, confusion matrices, precision, recall and F1-score, temporal topic modeling, emerging-threat detection, real-time detection analysis, error analysis, discussion, and limitations.

Chapter 6 – Conclusion and Future Work

This chapter summarizes the major findings and contributions of the research. It discusses the conclusions derived from the experimental results and presents potential directions for future research, including improved temporal modeling, larger CTI datasets, advanced language models, real-time data sources, and enhanced emerging-threat detection techniques.